

Clustering Applications and ML Pipeline Assignment

Clustering is an unsupervised machine learning technique used to group similar data points together. It is widely used in business analytics, healthcare, marketing, and recommendation systems.

Real-Life Clustering Applications

Application	Usage
Customer Segmentation	Grouping customers based on spending behavior
Fraud Detection	Detecting unusual transactions
Healthcare Analysis	Grouping patients with similar symptoms
Recommendation Systems	Movie and product recommendations
Retail Analysis	Grouping products bought together

Selected Applications

1. Synthetic 2D Data Grouping using K-Means Clustering

Dataset : make_blobs — 150 samples, 4 cluster centres

Features : Feature 1 and Feature 2

Method : Elbow Method to find optimal K, then KMeans(n_clusters=4)

2. Synthetic 2D Data Grouping using Hierarchical Clustering

Dataset : make_blobs — 150 samples, 3 cluster centres

Features : Feature 1 and Feature 2

Method : AgglomerativeClustering(n_clusters=3, linkage='ward')

K-Means Clustering Pipeline

A synthetic 2D dataset was generated using make_blobs with 150 samples and 4 cluster centres. Feature 1 and Feature 2 were selected as features. Data scaling was performed using StandardScaler. The Elbow Method was used to determine the optimal number of clusters. K-Means clustering grouped data points into 4 clusters. Silhouette Score was used for evaluation.

Hierarchical Clustering Pipeline

A synthetic 2D dataset was generated using make_blobs with 150 samples and 3 cluster centres. Agglomerative Clustering was applied using Ward linkage. Data points were grouped based on similarity in feature space. Silhouette Score was used to evaluate clustering quality.

Evaluation Metrics

K-Means Clustering

- Elbow Method was used to find the optimal number of clusters ($K = 4$)
- Silhouette Score was used to confirm cluster quality

Hierarchical Clustering

- Silhouette Score was used to evaluate clustering quality
- Data points were grouped based on similarity using Ward linkage

Business Metrics

- Better customer targeting
- Improved marketing strategy
- Increased customer retention
- Reduced marketing cost

Investor Benefits

- Higher ROI (Return on Investment)
- Better decision-making
- Increased profitability
- Smarter business growth strategies

Conclusion

Clustering techniques help in grouping similar data points without predefined labels. K-Means Clustering uses the Elbow Method to find the best K and assigns points to the nearest centroid. Hierarchical Clustering builds a tree of clusters using Ward linkage. Both models provide valuable insights for understanding data patterns, improving decision-making, and increasing profitability.